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A multimodal personality prediction framework based on adaptive graph transformer network and multi-task learning

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Abstract

Multimodal personality analysis targets accurately detecting personality traits by incorporating related multimodal information. However, existing methods focus on unimodal features while overlooking the bimodal association features crucial for this interdisciplinary task. Therefore, we propose a multimodal personality prediction framework based on an adaptive graph transformer network and multi-task learning. Firstly, we utilize pre-trained models to learn specific representations from different modalities. Here, we employ pre-trained multimodal models' encoders as the backbones of the modality-specific extraction methods to mine unimodal features. Specifically, we introduce a novel adaptive graph transformer network to mine personality-related bimodal association features. This network effectively learns higher-order temporal dependencies based on relational graphs and emphasizes more significant features. Furthermore, we utilize a multimodal channel attention residual fusion module to obtain the fused features, and we propose a multimodal and unimodal joint learning regression head to learn and predict scores for personality traits. We design a multi-task loss function to enhance the robustness and accuracy of personality prediction. Experimental results on the two benchmark datasets demonstrate the effectiveness of our framework, which outperforms the state-of-the-art methods. The code is available at https://github.com/RongquanWang/PPF-AGTNMTL.

CCS Concepts

• ***Imaging/Video*** → *Image/Video Processing;* • ***Interaction*** → *Multimodal/Cross-modal Interaction;* • ***Methods/Applications*** → *Artificial Intelligence/Machine Learning;*

1. Introduction

Personality computing is a subfield of affective computing that analyzes, recognizes, and synthesizes personality traits. Personality traits are persistent patterns of thoughts, feelings, and behaviors that shape and distinguish a person's personality from others [RK23]. Some researchers have utilized psychological scales to establish various personality trait models. For example, the Big Five model aims to assess personality through five characteristics: Openness (Open.), Conscientiousness (Cons.), Extraversion (Extr.), Agreeableness (Agre.), and Neuroticism (Neur.). With the blossoming of multimedia social platforms, such as YouTube, Instagram, and DouYin, people increasingly share their lives via videos [ZWL*23]. Meanwhile, CVPR organized the personality prediction competition in 2016 and 2017 based on videos collected from Youtube [GGB*17] and the First Impressions v2 dataset [EKS*18] released. These videos are annotated with apparent personality scores based on people's personality traits. Based on these facts, it is possible to automatically infer an individual's big five personalities from short videos. Personality traits greatly influence our lives and affect our behaviors, cognition, mental health, preferences, desires, and decisions. Besides, the ability to automatically detect one's personality traits has many important practical applications, such as job interviews [KGAS17], recommendation systems, social network analysis, personal assistants, purchasing behaviors, health care and counseling, criminology and so on. For example, when a company recruits a new employee, it can use automatic personality recognition technology to identify the interviewer's personality characteristics based on the interviewer's first impression to screen out more suitable employees.

Multimodal personality analysis aims to investigate the expressed personality traits in multimodal data [LLT22]. Because methods based on multimodal information can increase the robustness of predictions [SUS14], many personality prediction methods based on multimodal information have been proposed. Then, combining, filtering, and extracting required features from the data collected from different modalities has resulted in several methods emerging [LSG24]. Nevertheless, the inherent heterogeneity across modalities often introduces information redundancy and distribution gaps, increasing the difficulty of multimodal representation learning and feature fusion. Firstly, some methods based on

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visual information extract visual features through convolutional neural networks and 3D CNN architectures and use these features for personality prediction, such as [EBDOTA17], [HK20], [WSC*20], and [HSS18]. These methods mainly focus on visual features, ignoring audio and text features, and have limited utilization of multimodal data. Therefore, methods based on visual and audio information extract visual facial and scene features and audio features for personality prediction by designing two steam networks. These methods are currently the mainstream personality prediction methods, including [GGvGvL16], [SPM*16], [KGAS17], [VML17], [PPJE19], [SHZ*22], [SHL20], [MAMCMS20]. Meanwhile, we also introduce some audio-visual understanding methods based on other tasks for extracting audio and visual features and train predictive regression models for personality prediction tasks, such as [LSL*23] focuses on pretraining ViT and cross attention mechanisms, [GCDZ19] utilizes human hand and facial action features, [TLX20] uses unimodal and cross-modal temporal context, [HLTH24] is a long-form audio-visual video understanding method, [XZ22] introduces a cross-modal gated attention scheme that extracts visual features using global visual and audio signals, [GCX23] proposed a Joint Modal Mutual Learning (JML) process that adaptively dynamically calibrates various audible and visual information. Considering that text modalities also contribute to improving the accuracy of personality prediction, some methods based on visual, audio, and textual information have been proposed to achieve this goal by deploying different architectures, all of which aim to extract and integrate multimodal features through various spatial models, temporal models, and attention mechanisms, such as [ZLT*23], [RK23], and [GGB*17]. Specifically, these methods fuse multimodal features by attempting various feature fusion strategies, such as [WZZW17] and [ZZWW16] attempting early and late fusion, and [SSG*22] finding through experimental results that directly concatenate multimodal features yields better results, [WLFO23] designed an emotion guided multimodal fusion and contrastive learning framework for personality feature recognition. Meanwhile, some methods have also proposed new loss functions to guide the problem of inaccurate model training, such as [LWM*20]'s Bell Loss and [BLQ*24]'s Gaussian Loss. Additionally, some methods focus on the interpretability of personality prediction models, such as [EKS*18] and [EKS*20].

In recent years, pretraining has witnessed rapid development in multimodal processing [AYQ*21], especially for multimodal pretrained models [RLS*23], such as Contrastive Language-Image Pretraining (CLIP) [RKH*21] and a robust audio representation learning method by distilling from Contrastive Language-Image Pretraining (Wav2CLIP) [WSKB22], they not only can generalize well across domains and tasks without fine-tuning or task-specific architectures but also can provide embeddings corresponding to category labels, making them very suitable as feature extractors for personality prediction tasks with lacking data. For example, an interpretable personality prediction model based on CLIP [GSM23] has also been proposed. Furthermore, [SSJ*21] searched for the optimal CNN architecture, and the results were used to create a human-specific graph structure representation to identify the personality of the subject. This graph structure represents a novel end-to-end edge feature learning strategy that helps preserve the unique parameters of human-specific CNNs and the geometric relationships between their layers. Although various methods have been proposed, they extract and fuse unimodal features, ignoring multimodal association features. However, they have yet to fully utilize graph structures to explore the inherent relational information between different frame features regarding the effectiveness of multimodal fusion methods [RLDL24]. Existing personality prediction methods are based on single-task learning, so these methods cannot better explore the shared information and complementary features between different tasks [LS23, YXYW21]. Compared with single-task learning, multi-task learning can improve generalization ability and reduce the risk of model overfitting to enhance the robustness of the trained model. While previous work has focused on adjusting the weights between multi-tasks, ours is designed to learn unimodal and multimodal tasks with multi-task loss. It can learn similar information from multimodal tasks and differentiated information from unimodal tasks. Our framework can pay more attention to the differentiated representations of each unimodal itself by jointly training multimodal tasks and unimodal subtasks.

Given these challenges, we propose a novel multimodal personality analysis framework based on pre-trained models and an adaptive graph transformer network. Our framework combines one unimodal feature and two bimodal association features within a multi-task joint learning process, thereby modeling both the spatial-temporal features among modalities. We employ trained CLIP's text encoder, Bidirectional Gated Recurrent Unit (BiGRU) [HZXL21], Temporal Attention Block (TAB), and average operation derived from the audio transcript text to mine text feature for the unimodal feature. We utilize CLIP and Wav2CLIP to extract spatial text, visual, and audio features for bimodal association features. Then, the vector product and the proposed adaptive graph transformer network are used to learn longer temporal dependencies in unimodal sequence features. Finally, the cosine similarity and the average operation are utilized to extract the scene-descriptions and scene-audio association features. We first adopt a multimodal channel attention residual fusion module for the multimodal task to fuse features. Finally, we design a multi-task joint learning regression head with multi-task loss to train and predict scores for personality traits. Our main contributions can be summarized as follows:

- We propose a novel framework for multimodal personality prediction, which suggests an adaptive graph transformer network for extracting two bimodal association features, effectively capturing the temporal dependencies in intra-modal dynamics of non-verbal information.
- We design a multi-task joint learning regression head to learn personality-related cues effectively. Furthermore, we define a new multi-task loss function to enhance the robustness and accuracy of personality prediction.
- Extensive experiments on two benchmark datasets validate the effectiveness of our framework. Moreover, our framework achieves state-of-the-art performance.

2. Methodology

In this section, we elaborate on our proposed method with its diagram illustrated in Figure 1. The input of our framework is a short

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video, which mainly consists of a feature extraction module, multimodal channel attention residual fusion module, multi-task joint learning regression head module, and multi-task loss module. More details are introduced in the following subsections.

2.1. Feature extraction module

In this paper, we propose using CLIP and Wav2CLIP as feature extractors. We exploit their visual encoders to create embeddings for scene images and their text encoder or audio encoder to generate text or audio cue embeddings.

2.1.1. Scene-descriptions association feature using pre-trained CLIP

We design a scene-descriptions association feature extracting method based on CLIP [GSM23]. This method consists of three main steps: 1) generating embeddings for visual scene cues using Wav2CLIP's visual encoder; 2) generating embeddings of personality descriptions using CLIP's text encoder; 3) Calculating the similarity scores between the scene embeddings and descriptions embeddings by taking the vector product of them, and the dimensionality of features is reduced through average operation and MLP.

In this paper, we set N=15, and $N^{'}$= 1608. We use a list to produce embedding for the personality descriptions, which includes $N^{'}$=1608 adjectives that can accurately depict human personalities. This list consists of the personality terms carefully selected from personality surveys [CM08, Joh91] and multiple dictionaries [Dic89, Dic02, Mat09], ensuring the diversity and representativeness of descriptions to capture the complexity of human personality effectively. We then used CLIP's text encoder to generate description embeddings for these personality descriptions. Meanwhile, we used the "openai/clip-vit-large-patch14-336" model based on the ViT architectures [RKH*21] as Wav2CLIP's visual encoder to generate embeddings for the scene images from a video. Specifically, let $P = \{p_1, p_2, ..., p_{N'}\}$ denote the set of $N^{'}$ personality descriptions, and let $V = \{v_1, v_2, ..., v_N\}$ be a set of N scene images extracted from a video. For all personality descriptions P, we generate their embeddings TP using CLIP's text encoder. For all scene images V, we generate their embeddings SV using Wav2CLIP's visual encoder.

Using the scene images embeddings $SV \in R^{N\times 768}$ and personality description embeddings $TP \in R^{N^{'}\times 768}$, we calculated the similarity scores between them by taking the vector product of them and using them as feature vectors. We then calculate a feature matrix $F \in R^{N\times N^{'}}$. It is computed as follows:

$$F_{i,j} = SV_i TP_j^T \tag{1}$$

where each entry $F_{i,j}$ represents the similarity score between the i-th scene image and the j-th personality description. This resulted in a feature vector $N^{'}$ for each scene image, with each value corresponding to the similarity score between the scene image frame and the corresponding personality description.

Finally, we can learn a joint embedding space for scene images and personality descriptions, with scene images and their associated descriptions mapped to the same space. We then averaged the feature vector across all scene images for each video, resulting in a scene-descriptions association feature $F_{sd} \in R^{1\times N^{'}}$. This process is computed as follows:

$$F_{sd}^j = \frac{1}{N}\sum_{i=1}^{N} F_{i,j} \tag{2}$$

where F_{sd}^j is the feature value for the j-th personality description and the video and N is the number of frames in the video.

We construct and train an MLP regression model with two FC layers to reduce the dimension of the scene-descriptions association feature F_{sd}. Here, each FC layer consists of a linear layer, a BN layer, a ReLU activation layer, and a Dropout layer (parameter p=0.3), which reduces the dimensionality of features from 1608 to 512 and finally 128. Finally, we can obtain scene-descriptions association feature $f_{sd} \in R^{1\times 128}$.

2.1.2. Scene-audio association feature using pre-trained Wav2CLIP

As illustrated in Figure 2, to extract the scene-audio association feature, we first construct the scene-images relation sub-graph and audio-patches relation sub-graph, and then we propose an adaptive graph transformer network to aggregate the unimodal audio and scene features, respectively. Lastly, we use the similarity scores to obtain the scene-audio association feature. Below, we describe the extraction details of the feature:

2.1.2.1. Scene-images relation sub-graph. Based on the obtained the scene images feature matrix $SV \in R^{N\times 768}$. Then, we use sine and cosine functions to generate position encoding vectors for each scene image feature and add them to the original scene images feature $V^{'}$. To construct the scene-images relationship sub-graph $G_V = \{C_V, E_V\}$, where the nodes C_V aim to capture the semantics of different frame with scene images and the edges $\{E_V\}$ is used to measure the relation of different nodes (frames). Here, these visual features $V^{'}$ as the nodes $C_V = \{c_V^i\}_{i=1}^N \in R^{N\times d_V}$ of the scene-images relationship sub-graph, where c_V^i and d_V denotes the visual feature of the ith frame scene image and the dimension of the its visual feature. In this paper, all nodes in G_V are connected to each other, which means it is a complete graph.

2.1.2.2. Audio-patches relation sub-graph. This paper first uses the ffmpeg tool to convert videos into audio data for audio features, and the audio data is transformed into log-mel spectrogram. Next, we firstly divide the log-mel spectrogram into some patches, and each patch is individually fed into a Wav2CLIP's audio encoder for extracting patch feature, thereby obtaining the patch feature matrix $A^{'} \in R^{N\times 512}$, where N is the number of patches from the video. Without losing generality, the construction method of the audio-patches relation sub-graph is the same as that of the scene-images relation sub-graph, G_A.

2.1.2.3. Adaptive graph transformer network The adaptive graph transformer network proposed in this paper consists of two consecutive graph transformer modules. Meanwhile, each graph transformer module has three parts: position encoder, Graph Convolutional Network (GCN) convolution module, and transformer

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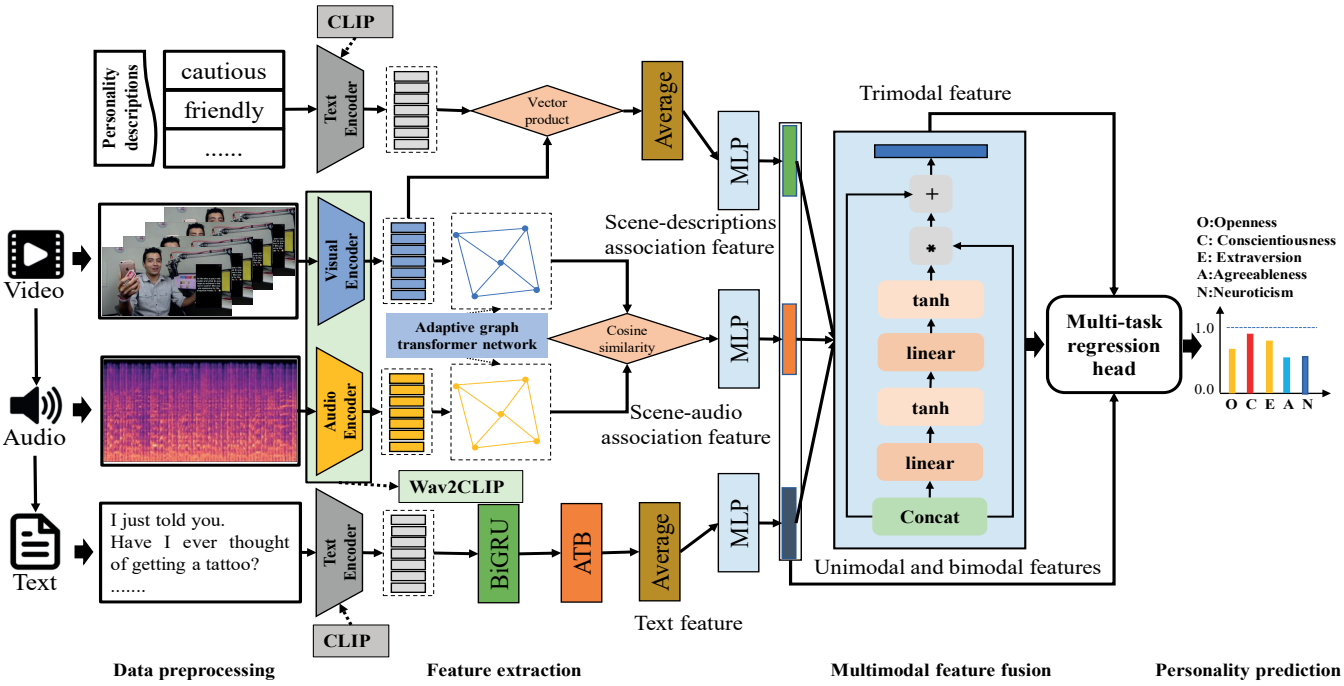

Figure 1: *The overall architecture of our framework. Our framework first employs CLIP's text encoder and Wav2CLIP to extract multimodal spatial features. Then, MLP and the proposed adaptive graph transformer network to learn the temporal relation in different unimodal, and the cosine similarity score is used to obtain the two association features from visual and audio information. Meanwhile, we utilize the multimodal channel attention residual fusion module for fusing features. Finally, a multi-task joint learning regression head was used to predict scores for personality traits.*

encoder. Each part of the graph transformer module is described as follows:

We first use a position encoder to generate position vectors PE for features and add them to the original features. Due to the fixed length of the data sequence, this paper uses the absolute position encoder in Transformer [VSP*17], which generates feature vectors at different positions using sine and cosine functions.

Secondly, we use a 1-layer standard GCN convolution to extract spatial features. The GCN convolution can be represented as:

$$X' = \hat{D}^{-\frac{1}{2}}\hat{A}\hat{D}^{-\frac{1}{2}}XW + b \tag{3}$$

where X is the inputs of scene-images feature matrix or audio-patches feature matrix, and X' are the outputs of GCN, $\hat{A}$ is an adjacency matrix with self-loops, $\hat{D}$ is the degree matrix of $\hat{A}$, and $\hat{D}_{ii} = \sum_{j=0}\hat{A}_{ij}$. W and b are GCN convolution weights and bias terms, respectively. Then, after GCN convolution, this paper uses BatchNorm regularization (BN), applies residual structure, and finally uses ReLU activation function (σ). This processing can be described as follows:

$$f_{GCN} = \sigma(X + BN(X')) \tag{4}$$

Next, we use a 1-layer standard transformer encoder layer to compute spatial-temporal features. The self-attention mechanism can be represented as:

$$Q,K,V = f_{GCN}W^Q, f_{GCN}W^K, f_{GCN}W^V \tag{5}$$

$$Attention(f_{GCN}) = softmax(\frac{QK^T}{\sqrt{d_k}})V \tag{6}$$

where Q, K, and V are the Query matrix, Key matrix, and Value matrix, respectively. W^Q, W^K, and W^V are the corresponding learnable weight parameters, and d_k represents the feature dimensions of Q and K. The multi-head attention mechanism can be expressed as:

$$MultiHead(f_{GCN}) = Concat(head_1, \ldots, head_i, head_h) \tag{7}$$

$$where\ head_i = Attention(f_{GCN}), i \in [1, \ldots, h], h = 8 \tag{8}$$

where $MultiHead(f_{GCN})$ is the transformer multi-head self-attention module, and this paper sets the number of attention heads for self-attention to $h = 8$ and applies dropout=0.1.

Finally, after outputting the structural features of the input graph through the self-attention module, this paper applies residual structure. It uses LayerNorm (LN) for regularization and further feature learning through two fully connected layers. Then, this paper uses residual structure and LayerNorm for regularization, as follows:

$$f_{MA} = LN(f_{GCN} + MultiHead(f_{GCN})), \tag{9}$$

$$f_{graph} = LN(f_{MA} + (\sigma(f_{MA}W_1 + b_1))W_2 + b_2) \tag{10}$$

where W_*, b_* are the weights and bias terms of the two fully connected layers.

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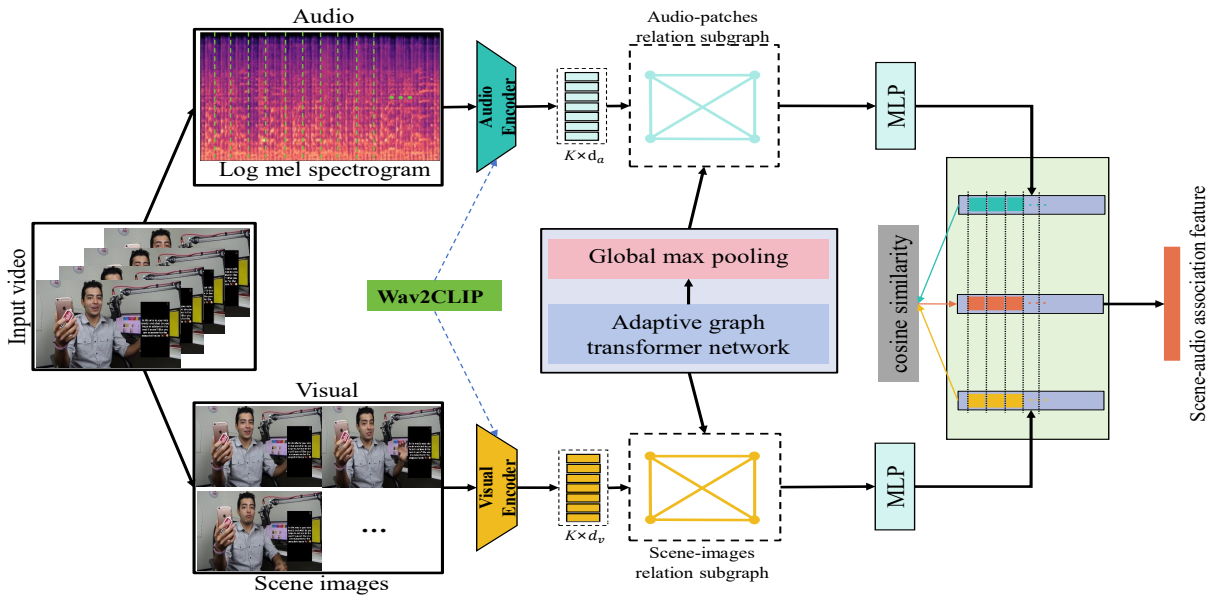

Figure 2: *Scene-audio association feature based on pre-trained Wav2CLIP and the designed adaptive graph transformer network.*

2.1.2.4. Scene-audio association feature. After constructing the scene-images relation sub-graph G_V and audio-patches relation sub-graph G_A, we first perform max pooling operations on both the two relation graphs to obtain scene-images relation sub-graph aggregation feature f_V and audio-patches relation sub-graph aggregation feature f_A, respectively. Finally, we can extract the scene-audio association feature by interacting with two sub-graph aggregation features with the cosine similarity, which utilizes both inner-modality and cross-modality of knowledge for personality prediction tasks. First, we extract scene-images relation sub-graph aggregation feature f_V. After feeding the new features of scene images into two consecutive graph encoders, the feature $f_{graph}^{V'} \in R^{N\times 768}$ of all scene images is obtained. Then, we use global max pooling aggregation to extract salient features $f_{graph}^{V} \in R^{1\times 768}$, the final feature representation $f_V \in R^{1\times 128}$ of the entire scene images relation graph is obtained using a fully connected layer, and it can be described as follows:

$$f_{graph}^{V'} = \sigma(f_{graph}^{V} W_V + b_V), \tag{11}$$

$$f_V = maxpool(f_{graph}^{V'}) \tag{12}$$

where W_V and b_V are the weights and bias terms of the feature mapping layer, and *maxpool* is the global maximum pooling.

Similar to aggregating scene-images relation sub-graph aggregation feature f_V, we can also aggregate the audio-patches relation sub-graph aggregation feature f_A using the adaptive graph transformer network and the global maximum pooling. Finally, this paper calculates the similarity score of f_V and f_A by cosine similarity function to obtain the scene-audio association feature $f_{va} \in R^{1\times 128}$ for scene images and audio patches. The entire calculation process is as follows:

$$f_{sa}^{t} = \frac{f_V^t f_A^t}{\sqrt{f_V^t + \epsilon}\sqrt{f_A^t + \epsilon}}, t = [1, 2, ..., 128] \tag{13}$$

where t represents the index of each feature vector between them, and ϵ is a minimum value of 10^{-8} added to prevent the dividend from being 0. Finally, we can obtain the scene-audio association feature f_{sa}.

2.1.3. Text feature

We first preprocess the input video for the text feature to obtain a transcript text. It is divided into several sentences; the maximum number of sentences is O, and features with fewer than O sentences are filled with zero vectors. Next, these sentences are fed into CLIP's text encoder to obtain text sequence features $X_T \in R^{O\times 768}$. Ultimately, by obtaining multiple sentence features from the video, a BiGRU [HZXL21] with TAB is utilized to extract temporal information and derive the text feature f_t. Finally, the dimensionality of text feature $f_t \in R^{1\times 128}$ is obtained through average operation and MLP. In this paper, we chose the BiGRU model because it performs better than other temporal models in extracting text features. For the experimental results and discussion analysis, please refer to section 7. Effectiveness of different temporal models in terms of text feature in Additional Material, O= 13 in CFI-V2 dataset and O= 452 in UDIVA v0.5 dataset.

2.2. Multimodal channel attention residual fusion module

Existing fusion methods directly concatenate multiple features, ignoring that different features affect personality prediction differently. Therefore, a multimodal channel attention residual fusion module is used to obtain trimodal feature. Specifically, the three feature vectors acquired are concatenated into a fused feature vector, which is then used as input for the module to evaluate the

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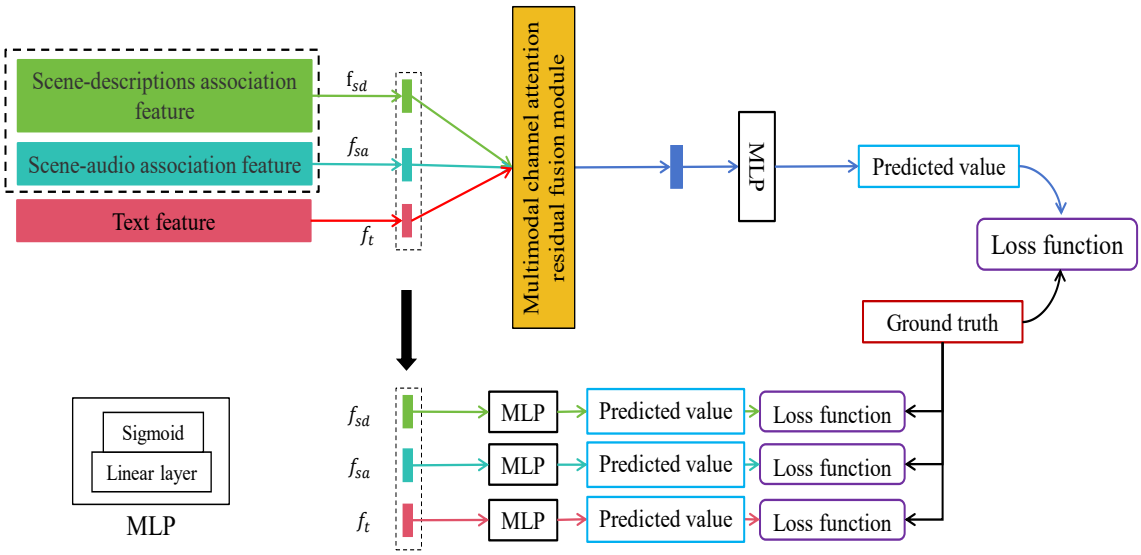

Figure 3: *The structural of multimodal and unimodal joint learning regression head. The MLP model consists of a linear layer and a sigmoid activation layer, which map the fused feature to 5 dimensions and normalize the output values to between 0 and 1, representing the prediction value of the Big Five personality.*

contribution of each branch and minimize information redundancy arising from feature diversity. A residual structure is incorporated into this module to mitigate information loss. The module comprises two fully connected layers to calculate the attention weight α for each dimension in the multimodal representation F, α can be calculated as:

$$\alpha = tanh(W_2 tanh(W_1 F + b) + c) \tag{14}$$

where W_1, W_2, b, and c represent the weight matrices and bias of the two fully connected layers, respectively. *tanh* is used to confine the attention weight within the interval $[-1, 1]$, and then the obtained attention vector α performs element multiplication with each dimension of the multimodal vector F. The formula is as follows:

$$F^{'} = F \cdot \alpha + F \tag{15}$$

where $F^{'}$ is the module's output, it is fed into the regression head to predict the five personality traits.

2.3. Multi-task joint learning regression head

We mainly introduce a multi-task joint learning regression head to predict the scores of five personality traits. Here, the three individual unimodal tasks (f_{sd}, f_{sa}, f_t) share modal representations with the multimodal task is illustrated in Figure 3. It adopts a multi-modal channel attention residual fusion module to obtain the fused feature. Afterward, the fused feature is processed through a single-layer MLP model to predict the scores for personality traits.

To achieve more robust personality prediction accuracy, this paper uses real Big Five personality ground truths for supervised learning of the output of multimodal prediction heads. It achieves auxiliary supervised learning of our framework by adding auxiliary prediction heads to every single feature. Among them, the MLP model is added to each single modality to complete the personality trait prediction output of the single modality. The MLP model consists of linear and sigmoid activation layers for all single features, with a feature mapping output dimension 5. It is worth noting that all single-modal auxiliary prediction heads are only used to assist prediction model training. Eventually, we jointly learn multi-modal and unimodal tasks. Notably, these unimodal tasks are solely present during the training stage to improve the robustness and accuracy of the model.

2.4. Multi-task loss module

We jointly optimize the entire framework with a multi-task loss function, and it can be defined as follows:

$$\begin{aligned} a &= Loss_{LogCosh} + Loss_{Bell} \\ Loss_{Total} &= \frac{1}{N}\sum_{i}^{N}(a + \alpha \sum_{s}^{\{VT_{sd}, VA_{sa}, T_e\}} a) \end{aligned} \tag{16}$$

where $Loss_{logCosh}$ is logCosh loss, and $Loss_{Bell}$ is the Bell Loss. N is the number of training samples. α is the importance of unimodal auxiliary task s for multi-modal task loss. In this paper, we set α as 0.2, and then $s = \{f_{sd}, f_{sa}, f_t\}$ represent different unimodal auxiliary tasks, respectively.

3. Experiment

3.1. Datasets and Metrics

In this paper, the publicly available ChaLearn First Impression-V2(CFI-V2) [EGE*17] dataset and the multilingual UDIVA v0.5 dataset (https://chalearnlap.cvc.uab.es/dataset/41/description/) [PSS*21] are utilized for personality prediction. The CFI-V2 dataset comprises 10,000 videos (average duration: 15s), and it

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Table 1: *Comparison with state-of-the-art methods on two benchmark datasets.*

Datasets→	CFI-V2				UDIVA v0.5			
Methods↓	ACC	PCC	CCC	R^2	ACC	PCC	CCC	R^2
Ventura, Masip, and Lapedriza. 2017 [VML17]	0.9116	0.5922	0.5402	0.3495	0.8484	0.0170	-0.0004	-0.1033
Girdhar et al. 2019 [GCDZ19]	0.8798	0.0221	0.0004	-0.0231	0.8473	-0.0271	-0.0009	-0.0597
Wang et al. 2020 [WSC*20]	0.8813	0.0492	0.0078	0.0013	0.8505	0.1541	0.0257	-0.0722
Helm and Kampel. 2020 [HK20]	0.8940	0.4188	0.3227	0.1756	0.8471	0.0514	0.0000	-0.1306
Suen, Hung, and Lin. 2020 [SHL20]	0.9064	0.6150	0.5925	0.3590	0.8419	0.1244	0.0343	-0.1326
Moreno-Armend´ariz et al. 2020 [MAMCMS20]	0.9058	0.6069	0.5852	0.3455	0.8485	0.1502	0.0545	-0.0425
Subramaniam et al. 2016 [SPM*16]	0.9044	0.5801	0.5459	0.3239	0.8470	0.0596	0.0000	-0.1316
Güçlütürk et al. 2016 [GGvGvL16]	0.9110	0.6428	0.6227	0.3758	0.8316	0.1482	0.1228	-0.2032
Hu et al. 2018 [HSS18]	0.8869	0.2891	0.1490	0.0845	0.8428	0.1522	0.1214	-0.1742
Suman et al. 2022 [SSG*22]	0.9140	0.6747	0.6292	0.4578	0.8484	0.0451	0.0059	-0.0798
Sun et al. 2022 [SHZ*22]	0.8956	0.4494	0.3697	0.2024	0.8463	-0.0164	-0.0089	-0.0938
Ryumina et al. 2023 [RK23]	0.9180	0.6729	0.6432	0.4539	0.8489	0.1990	0.0418	-0.0874
Bao et al. 2024 [BLQ*24]	0.8957	0.4497	0.3534	0.2017	0.8494	-0.0193	0.0000	-0.0760
Li et al. 2020 [LWM*20]	0.9167	0.6829	0.6478	0.4897	0.8468	0.2200	0.1326	-0.0320
Lin et al., 2023 [LSL*23]	0.8959	0.4652	0.3835	0.2135	0.8486	0.1382	0.0119	-0.0904
Tian et al., 2020 [TLX20]	0.9129	0.6703	0.6422	0.4433	0.8480	0.2515	0.0870	-0.0388
Hou et al., 2024 [HLTH24]	0.9123	0.6614	0.6063	0.4349	0.8468	0.1979	0.0287	-0.0373
Xia and Zhao, 2022 [XZ22]	0.9122	0.6638	0.6277	0.4307	0.8366	0.2521	0.0872	**0.1340**
Gao et al., 2023 [GCX23]	0.9163	0.6957	0.6465	0.4855	0.8508	0.2861	0.0356	-0.0680
Gan et al., 2023 [GSM23]	0.9176	0.6550	0.6292	0.4225	0.8479	0.2578	0.2178	-0.0695
Shao et al., 2021 [SSJ*21]	0.9183	0.6638	0.6457	0.4329	0.8521	0.1426	0.0907	-0.0326
Our framework	**0.9196**	**0.7267**	**0.7085**	**0.5226**	**0.8583**	**0.4211**	**0.2289**	0.1043

Note that the values in bold denote the highest ones in the columns, and the values underlined denote the second-highest ones.

provides official splits where the training, validation, and test sets contain 6000, 2000, and 2000 videos, respectively. The UDIVA v0.5 dataset, a preliminary subset of the UDIVA dataset [PSS*21], is officially divided into three subject-independent splits: 116/18/11 sessions and 99/20/15 participants in training, validation, and test, respectively. Please refer to section 1. Datasets in Additional Material for more details.

To measure the performance of our framework, we first employ four standard evaluation metrics i.e., the mean accuracy (ACC), the Pearson Correlation Coefficient (PCC), the Concordance Correlation Coefficient (CCC), and the coefficient of determination (R^2) proposed in [LSG24, LWM*20, PSS*21] and [SJS*21, GZVB*23]. Please refer to section 2.Evaluation metrics in Additional Material for more details.

3.2. Implementation Details

We conducted our experiments on a server with Pytorch consisting of a 2.90GHz Intel(R) Xeon(R) Gold 6226R CPU with 24GB of RAM and one NVIDIA GeForce RTX 3090 GPU.For more implementation details of our framework, please refer to the section 3. Implementation Details in Additional Material.

3.3. Results and Analysis

In this section, the proposed framework's performance is evaluated in some experiments. First, the effectiveness of our framework is verified by comparing it with other methods. Then, we investigate the effectiveness of different features and components. Finally, we perform thorough case studies to ascertain the performance of our framework.

3.3.1. Comparison with the state-of-the-art methods

In this section, we compare our framework with several state-of-the-art methods on the CFI-V2 and UDIVA v0.5 datasets to demonstrate the efficacy of our framework. On the one hand, we comprehensively evaluate the performance of our framework and other comparative methods using four metrics. The four metric scores of different state-of-the-art methods are shown in Table 1. The significant findings can be summarized as follows: First, except for the value of R^2 in UDIVA v0.5, our framework achieves the highest accuracy for all other metrics in two benchmark datasets. Second, in terms of PCC and CCC, our framework's accuracy on both datasets is much higher than other excellent comparison methods, indicating that our framework has a strong personality prediction ability, and the difference between our framework's predicted values and true values is small. Therefore, the experimental results demonstrate the superiority of our framework in personality assessment.

On the other hand, to further demonstrate the excellent performance of our framework on two datasets, we also directly cited the ACC of comparison methods from their paper for comparison based on CFI-V2. Please refer to section 4. Comparison with the state-of-the-art methods based on CFI-V2 dataset in Additional Material. Meanwhile, we also evaluated the performance of our framework on 4 tasks of the UDIVA v0.5 dataset, please refer to section 5 in Additional Material. Finally, to understand the performance of different features on this framework and their impact on our framework, we also conducted corresponding ablation experiments to demonstrate it, please refer to section 6. Performance and effectiveness of different features in Additional Material.

Based on the above results, our framework outperforms all the compared methods and produces the most competitive performance on two benchmark datasets.

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Table 2: *Results of different temporal models on the two benchmark datasets.*

Datasets→	CFI-V2				UDIVA v0.5			
Metrics→	ACC	PCC	CCC	R^2	ACC	PCC	CCC	R^2
RNN	0.9196	0.7252	0.7045	**0.5239**	0.8551	0.3167	0.1919	0.0585
BiRNN	0.9194	0.7228	0.7046	0.5191	0.8528	0.3411	0.2208	0.0513
GRU	0.9194	0.7245	0.7077	0.5203	0.8527	0.3291	0.2384	0.0401
BiGRU	0.9192	0.7203	0.6987	0.5166	0.8554	0.3580	0.2111	0.0704
LSTM	0.9192	0.7215	0.7029	0.5168	0.8531	0.3455	0.1434	0.0145
BiLSTM	0.9191	0.7220	0.7040	0.5164	0.8553	0.4175	0.1424	0.0400
Our adaptive graph transformer network	**0.9196**	**0.7267**	**0.7085**	0.5226	**0.8583**	**0.4211**	**0.2289**	**0.1043**

Table 3: *Comparison on the performance of our framework with different losses, where "w/o" means removal of the loss from our framework.*

Datasets→	Loss functions	CFI-V2				UDIVA v0.5			
Metrics→		ACC1	PCC	CCC	R^2	ACC2	PCC	CCC	R^2
	LogCosh loss only	**0.9198**	0.7255	0.7048	**0.5246**	0.8537	0.2997	0.1928	0.0431
w/o Our multi-task loss	Bell loss only	0.9195	0.7243	0.7050	0.5219	0.8570	0.3822	0.2085	0.0809
	LogCosh+Bell losses	0.9195	0.7250	0.7044	0.5218	0.8580	0.3946	0.1870	0.0986
Our multi-task loss	LogCosh+Bell losses	0.9196	**0.7267**	**0.7085**	0.5226	**0.8583**	**0.4211**	**0.2289**	**0.1043**

3.3.2. Effectiveness of our adaptive graph transformer network

Existing temporal models, such as RNNs, commonly suffer from problems such as vanishing/exploding gradients and a limited ability to learn long-term dependencies. However, our designed adaptive graph transformer network is not limited by the temporal relationships between frames. It can utilize the topological structure of the complete graph to transfer and aggregate features, thereby capturing long-distance dependencies between different frames. Based on the experiment results of Table 2, our adaptive graph transformer network is used to mine two bimodal association features, and it achieved relatively excellent performance in most metrics on both datasets compared to other temporal models.

3.3.3. Effectiveness of our multi-task loss function

To verify the effectiveness of our multi-task loss. We further compared the performance of our framework by training it with different losses, including Log-Cosh and Bell losses, as well as their combination loss and our multi-task loss. Our framework served as the backbone model, with only the loss function changed for this comparison. As shown in Table 3, when all two losses are used together, the performance is slightly improved as well, benefiting from the complementary nature of the two losses. Furthermore, our framework with multi-task loss has achieved significant improvements compared to combined two losses, which confirms the effectiveness of our multi-task loss for multimodal personality analysis.

3.3.4. Case study

To qualitatively validate the effectiveness of our framework, we display four typical cases in Figure 4, where personality traits are assessed by our framework and three SOTA methods, Gao et al., 2023 [GCX23], Shao et al., 2021 [SSJ*21], and Li et al. 2020 [LWM*20]. These comparison methods did not utilize the proposed association features and multi-task learning for personality trait prediction. However, our framework can accurately predict various personality trait scores using them. The results of these cases (i.e., cases A and B in two datasets) show our framework outperforms these SOTA methods in performance. Specifically, the five personality score values predicted by our framework are closer to the true values compared to those predicted by other comparison SOTA methods, while the predicted values of other comparison SOTA methods have greater differences compared to the true values. Moreover, these cases are a microcosm of most cases, indicating that our framework can accurately predict these personality traits in various situations.

4. Conclusion

In this paper, we have presented a novel multimodal feature learning framework for predicting personality, which involves the scene-audio association feature, the scene-descriptions association feature, and a multi-task joint learning regression head with a multi-task loss to improve the performance of personality prediction effectively. Extensive experiments validate the effectiveness and reliability of our framework. The results show that our framework outperforms the current state-of-the-art approaches. For future work, we plan to explore self-supervised learning methods and psychological clues to improve the performance of our framework and apply it to real-world applications such as job interviews.

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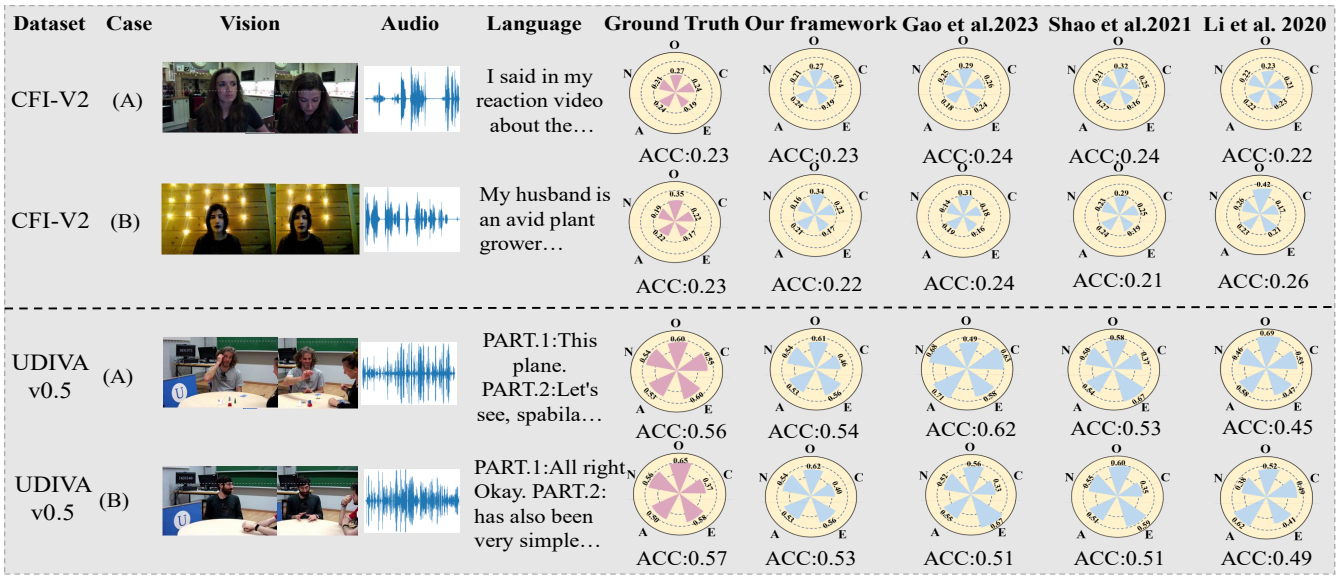

Figure 4: *Four cases with Big-Five personality traits assessed by our framework, Gao et al., 2023 [GCX23], Shao et al., 2021 [SSJ*21], and Li et al.2020 [LWM*20].*

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